The Neumann stochastic square in the planar CH/AC homotopy

Candidate solution packet for arXiv:2204.09545

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Status. Candidate full answer for smooth bounded planar domains, likely valid, awaiting specialist review.

1 Source problem

Blömker and Tölle isolate the following missing estimate in Section 6 of [1], on page 18 of the source PDF.

6 Stochastic Convolution for the CH/AC-homotopy

In order to finish the full approximation result, we need to establish various bounds on the stochastic convolution. We shall briefly state the corresponding results on the flat torus only, i.e. for periodic boundary conditions. We assume moreover that the operator Q_ε is Fourier-diagonal and hence is equal to the operator square root of the covariance operator $Q_\varepsilon^* Q_\varepsilon = Q_\varepsilon^2$.

For Neumann boundary conditions on a general domain in particular the quantification of the convergence for Z_ε^2 seems to be an open problem. So far, we are only aware of the result [21] with Neumann boundary conditions on the square.

On the two-dimensional torus, we can rely on Fourier series. Assuming that Q_ε is also diagonal in Fourier space, we have

$$Z_\varepsilon(t) = \sum_{k\in\mathbb{Z}^2} \alpha_k(\varepsilon) I_k^{(\varepsilon)}(t) e_k \quad \text{with} \quad I_k^{(\varepsilon)}(t) = \int_0^t e^{-(t-s)\lambda_k(\varepsilon)}\, d\beta_k(s)$$

We prove the requested convergence for every connected bounded C^∞ domain in $\mathbb{R}^2$. The proof also quantifies the exact noise scaling and limiting constant.

2 Statement

Let $L = -\Delta_N$ be the nonnegative Neumann Laplacian on $L^2(\Omega)$. Write

$$Le_j = \mu_j e_j, \qquad e_0 = |\Omega|^{-1/2}, \quad \mu_0 = 0,$$

and work first on the mean-zero subspace, so indices below start at 1. For $0 < \varepsilon < 1/2$, put

$$a_\varepsilon = 1 - \varepsilon, \qquad B_\varepsilon = L(a_\varepsilon + \varepsilon L), \qquad \lambda_j^\varepsilon = \mu_j(a_\varepsilon + \varepsilon\mu_j).$$

If W_0 is cylindrical white noise on the mean-zero subspace, define the zero-initial stochastic convolution

$$Z_\varepsilon(t) = \sigma_\varepsilon \int_0^t e^{-(t-s)B_\varepsilon}\, dW_0(s).$$

For fixed ε, the fourth-order regularization makes the ordinary square Z_ε^2 well-defined.

We use the Bessel-potential norm $\|f\|_{H^{-1}} = \|(I+L)^{-1/2} f\|_2$.

Theorem 1. *Suppose*

$$\sigma_\varepsilon^2 \log(1/\varepsilon) \longrightarrow \kappa \quad \textit{for some } \kappa \in [0,\infty).$$

Then for every $T<\infty$, $1 \le p < \infty$, and $q \ge \max\{2,p\}$,

$$\left(\mathbb{E}\left[\left\| Z_\varepsilon^2 - \frac{\kappa}{8\pi}\right\|^q_{L^p(0,T;H^{-1}(\Omega))}\right]\right)^{1/q} \longrightarrow 0.$$

More precisely, with constants depending on Ω, T, p, q but not on ε,

$$\left(\mathbb{E}\left[\left\| Z_\varepsilon^2 - \frac{\kappa}{8\pi}\right\|^q_{L^p H^{-1}}\right]\right)^{1/q} \le C_{p,q,T}\left(\left|\sigma_\varepsilon^2 \log(1/\varepsilon) - \kappa\right| + \sigma_\varepsilon^2\right). \tag{1}$$

In particular the convergence holds in probability in every finite $L^p(0,T;H^{-1})$.

The source suppresses the constant Fourier mode on the torus. The same convention is used here. Restoring the one-dimensional Neumann kernel adds terms that are $O(\sigma_\varepsilon)$ in the relevant Gaussian norms and does not alter the conclusion.

3 Proof intuition

Although products of Neumann eigenfunctions have no Fourier convolution law, the linear operator is still a function of the Neumann Laplacian. Its covariance is therefore an integral of the ordinary Neumann heat kernel. On the diagonal, the interior heat kernel contributes the universal $(8\pi)^{-1}\log(1/\varepsilon)$ divergence. The reflected boundary term is large only in a shrinking collar and leaves an L^2-bounded remainder.

After subtracting the mean, Wick's formula turns the H^{-1} variance of the square into the integral of three kernels: one Bessel Green kernel and two covariance kernels. Each has only a logarithmic diagonal singularity, and a logarithmic cube is integrable in two dimensions. The remaining factor is σ_ε^4, so the random part vanishes. This replaces the torus-specific Fourier cancellation by a domain-independent heat-kernel argument.

4 Two heat-kernel estimates

Let $p_s(x,y)$ be the Neumann heat kernel and $p_s^0(x,y) = p_s(x,y) - |\Omega|^{-1}$. We use two standard consequences of the Gaussian estimate and the not-feeling-the-boundary estimate [2]. If $d(x) = \mathrm{dist}(x,\partial\Omega)$, then, for $0 < s \le 1$,

$$|p_s(x,y)| \le Cs^{-1}e^{-|x-y|^2/(Cs)}, \tag{2}$$

$$\left|p_s(x,x) - (4\pi s)^{-1}\right| \le Cs^{-1}e^{-d(x)^2/(Cs)} + Cs^{-1/2}. \tag{3}$$

The second term is the harmless lower-order parametrix remainder. For $s \geq 1$, p_s^0 decays exponentially by the spectral gap.

Define the unit-amplitude stationary covariance kernel

$$Q_\varepsilon(x,y) = \sum_{j\geq 1} \frac{e_j(x)e_j(y)}{2\lambda_j^\varepsilon}. \tag{4}$$

Lemma 1. *Uniformly for $0 < \varepsilon < 1/2$,*

$$Q_\varepsilon(x,x) = \frac{1}{8\pi}\log(1/\varepsilon) + R_\varepsilon(x), \qquad \sup_\varepsilon \|R_\varepsilon\|_{L^2(\Omega)} < \infty, \tag{5}$$

$$|Q_\varepsilon(x,y)| \leq C(1 + |\log|x-y||), \qquad x \neq y. \tag{6}$$

The kernel G_1 of $(I+L)^{-1}$ obeys the same off-diagonal logarithmic bound and is nonnegative.

Proof. For $\mu > 0$, partial fractions and the Laplace transform give

$$\frac{1}{\mu(a_\varepsilon + \varepsilon\mu)} = \frac{1}{a_\varepsilon}\int_0^\infty (1 - e^{-a_\varepsilon s/\varepsilon})e^{-\mu s}\, ds.$$

Consequently,

$$Q_\varepsilon(x,y) = \frac{1}{2a_\varepsilon}\int_0^\infty (1 - e^{-a_\varepsilon s/\varepsilon})p_s^0(x,y)\, ds. \tag{7}$$

Estimate (2), followed by the change of variables $u = |x-y|^2/s$, proves (6). The same heat representation, without the cutoff factor, proves the assertion for G_1.

On the diagonal, substitute $(4\pi s)^{-1}$ in the part of (7) with $0 < s < 1$. Since $a_\varepsilon \in [1/2, 1]$,

$$\frac{1}{8\pi a_\varepsilon}\int_0^1 (1 - e^{-a_\varepsilon s/\varepsilon})\frac{ds}{s} = \frac{1}{8\pi}\log(1/\varepsilon) + O(1).$$

The large-time and constant-mode terms are uniformly bounded. By (3), the absolute value of the remaining boundary term is bounded by a constant plus

$$H_\varepsilon(d(x)) := \int_0^1 \min\{1, s/\varepsilon\}\, s^{-1} e^{-d(x)^2/(Cs)}\, ds.$$

For $d \leq \sqrt{\varepsilon}$, this is at most $C\log(1/\varepsilon)$; for $d > \sqrt{\varepsilon}$, it is at most $C(1 + |\log d|)$. Smooth boundary normal coordinates therefore give

$$\sup_{\varepsilon < 1/2} \int_\Omega H_\varepsilon(d(x))^2\, dx < \infty,$$

because $\sqrt{\varepsilon}\log^2(1/\varepsilon)$ is bounded and $\int_0^1 |\log r|^2 dr < \infty$. This proves (5). $\square$

5 The deterministic mean

Set $m_\varepsilon(t,x) = \mathbb{E}Z_\varepsilon(t,x)^2$. Spectral calculus gives

$$m_\varepsilon(t,x) = \sigma_\varepsilon^2 Q_\varepsilon(x,x) - D_\varepsilon(t,x), \quad D_\varepsilon(t,x) = \sigma_\varepsilon^2 \sum_{j\geq 1} \frac{e^{-2t\lambda_j^\varepsilon}}{2\lambda_j^\varepsilon} e_j(x)^2. \tag{8}$$

Lemma 1 shows

$$\left\|\sigma_\varepsilon^2 Q_\varepsilon(\cdot,\cdot)|_{x=y} - \frac{\kappa}{8\pi}\right\|_{H^{-1}} \le C\left(|\sigma_\varepsilon^2 \log(1/\varepsilon) - \kappa| + \sigma_\varepsilon^2\right). \tag{9}$$

Since $\lambda_j^\varepsilon \ge \mu_j/2$,

$$0 \le \frac{D_\varepsilon(t,x)}{\sigma_\varepsilon^2} \le \sum_{j\ge 1} \frac{e^{-t\mu_j}}{\mu_j} e_j(x)^2 = \int_t^\infty p_s^0(x,x)\,ds \le C_T(1 + |\log t|),$$

where the final bound follows from (2) and the spectral gap. Thus, for every finite p,

$$\left\|m_\varepsilon - \frac{\kappa}{8\pi}\right\|_{L^p(0,T;H^{-1})} \le C_{p,T}\left(|\sigma_\varepsilon^2 \log(1/\varepsilon) - \kappa| + \sigma_\varepsilon^2\right). \tag{10}$$

6 The centered square

Let $Y_\varepsilon(t) = Z_\varepsilon(t)^2 - m_\varepsilon(t)$. First truncate the Neumann expansion at a finite index. Wick's formula gives

$$\mathbb{E}\|Y_\varepsilon(t)\|_{H^{-1}}^2 = 2\sum_{i,j\ge 1} c_i(t)c_j(t)\left\langle e_ie_j, (I+L)^{-1}(e_ie_j)\right\rangle, \tag{11}$$

where

$$c_j(t) = \frac{\sigma_\varepsilon^2(1-e^{-2t\lambda_j^\varepsilon})}{2\lambda_j^\varepsilon} \le \frac{\sigma_\varepsilon^2}{2\lambda_j^\varepsilon}.$$

Every summand in (11) is nonnegative. Hence monotone passage from finite truncations and the kernel identity yield

$$\begin{aligned}\mathbb{E}\|Y_\varepsilon(t)\|_{H^{-1}}^2 &\le 2\sigma_\varepsilon^4 \iint_{\Omega^2} G_1(x,y) Q_\varepsilon(x,y)^2\,dx\,dy \\ &\le C\sigma_\varepsilon^4.\end{aligned}$$

The last integral is uniformly finite by Lemma 1, since $(1+|\log|x-y||)^3$ is locally integrable in two dimensions.

The random variable $Y_\varepsilon(t)$ takes values in the second homogeneous Wiener chaos. Vector-valued Gaussian hypercontractivity therefore gives, for every $q \ge 2$,

$$\left(\mathbb{E}\|Y_\varepsilon(t)\|_{H^{-1}}^q\right)^{1/q} \le C_q\sigma_\varepsilon^2.$$

For $q \ge p$, Minkowski's integral inequality now implies

$$\left(\mathbb{E}\|Y_\varepsilon\|_{L^p(0,T;H^{-1})}^q\right)^{1/q} \le C_{p,q,T}\sigma_\varepsilon^2. \tag{12}$$

Combining (10) and (12) proves Theorem 1.

7 Scope, verification, and novelty

The theorem supplies precisely the domain-dependent estimate isolated in the source paper, for arbitrary smooth shape rather than a square or torus. It also clarifies that the nontrivial regime is expressed invariantly as $\sigma_\varepsilon^2 \log(1/\varepsilon) \to \kappa$; whether a paper denotes noise amplitude or variance by σ_ε can otherwise obscure a square root.

The smooth-boundary hypothesis is used only in the boundary-collar form of the heat-kernel estimate. We make no claim here for rough domains or colored noise operators that fail to commute with L. A numerical companion checks the coefficient $1/(8\pi)$ from Neumann eigenvalues on the unit square; it is not part of the proof.

A bounded novelty search through 13 August 2026 covered the run registry, local full sources, the exact open phrase, arXiv:2204.09545, and web searches combining CH/AC homotopy, Neumann boundary, general domain, stochastic convolution, and Wick square. No later explicit resolution was found. This is not an exhaustive priority search.

Human-review recommendation. Verify first Lemma 1, especially the uniform L^2 treatment of the boundary collar. Next verify the nonnegative spectral expansion (11) and its kernel identity. These are the only substantive points; the remaining bounds are direct.

References

[1] D. Blömker and J. M. Tölle, *Singular limits for stochastic equations*, Stochastics and Dynamics (2023), DOI: 10.1142/S0219493723500405; arXiv:2204.09545.

[2] L. Li and A. Strohmaier, *Heat kernel estimates for general boundary problems*, J. Spectr. Theory 6 (2016), 903–919; arXiv:1604.00784.